Causality Assessment of Adverse Drug Reactions: Methods and Case Classification

Rhabdomyolysis after clarithromycin added to simvastatin (classified Probable)

Scenario:

A 68-year-old woman (58 kg) with hypertension and dyslipidemia took simvastatin 40 mg HS (3 years), amlodipine 5 mg OD (3 years) and hydrochlorothiazide 12.5 mg OD (2 years). Baseline creatinine was 0.9 mg/dL and she had no muscle complaints. A physician unaware of her statin therapy prescribed clarithromycin 500 mg BD for 7 days for a lower respiratory tract infection. On day 3 she had mild thigh and calf pain, and by day 4 she had marked proximal muscle weakness, difficulty climbing stairs and cola-colored urine. She was admitted on day 6, and clarithromycin and simvastatin were both stopped. She received IV fluids with electrolyte monitoring, CK fell by day 10, and by day 14 creatinine was 1.0 mg/dL and the weakness had resolved.

Admission findings (day 6): CK 18,500 U/L (reference <170), creatinine 1.9 mg/dL, potassium 5.6 mmol/L, AST 420 U/L, ALT 210 U/L, urine myoglobin positive, TSH normal. Trauma, strenuous exercise, alcohol use and seizures were excluded.

Assessment:

- **Time relationship:** Symptoms began within 3 to 4 days of adding clarithromycin, after three stable years on simvastatin, which fits a newly introduced interacting drug.
- **Suspect drugs:** Clarithromycin is the precipitant, a strong CYP3A4 and P-gp inhibitor. Simvastatin is the object drug, a CYP3A4 substrate whose exposure can rise several-fold. Amlodipine is a weak CYP3A4 inhibitor, and simvastatin should not exceed 20 mg/day with it, so her 40 mg dose was already above the recommended cap. Hydrochlorothiazide is unlikely.
- **Alternative causes:** None found. The raised AST and ALT reflect release from muscle and not primary liver disease. Hypothyroidism was excluded by the normal TSH.
- **Dechallenge:** Both drugs were stopped together, so recovery supports a drug cause but cannot separate the two drugs. This is why the interacting pair is assessed as the cause.
- **Rechallenge:** Not performed, and it should not be attempted.
- **Risk factors:** Age over 65, female sex, low body weight and polypharmacy.

Naranjo score:

#	Questions	Answer	Score
1	Are there previous conclusive reports on this reaction?	Yes (known interaction and reaction)	+1

#	Questions	Answer	Score
2	Did the adverse event occur after the suspected drug was given?	Yes	+2
3	Did the adverse reaction improve when the drug was discontinued or a specific antagonist was given?	Yes	+1
4	Did the adverse reaction reappear when the drug was re-administered?	Not done	0
5	Are there alternative causes (other than the drug) that could on their own have caused the reaction?	No alternative causes	+2
6	Did the reaction reappear when a placebo was given?	Not done	0
7	Was the drug detected in blood (or other fluids) in concentrations known to be toxic?	Drug levels not measured	0
8	Was the reaction more severe when the dose was increased, or less severe when the dose was decreased?	Not assessed	0
9	Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	No	0
10	Was the adverse event confirmed by any objective evidence?	Yes (CK, myoglobinuria)	+1
		Total	**7**

Result: Naranjo **Probable**; WHO-UMC **Probable**.

Why not Certain: There was no rechallenge, no measured drug level, and two drugs were withdrawn at the same time. Because this is a drug-drug interaction, the assessment applies to the combination of clarithromycin and simvastatin and not to either drug alone.

Related classifications (separate from causality):

- **Severity (Hartwig):** Level 4b, moderate, because the reaction caused hospital admission.
- **Seriousness:** Serious, because of hospitalization and acute kidney injury.
- **Preventability (Schumock-Thornton):** Preventable. A known, labeled interaction was overlooked, the medication history was not checked, and the simvastatin dose exceeded the amlodipine-related cap.

Management points:

Stop both suspected drugs, give aggressive IV hydration, and monitor CK, creatinine, potassium and urine output. After recovery, use a statin that does not depend on CYP3A4, such as pravastatin or rosuvastatin. If a macrolide is needed in future, azithromycin is a much weaker CYP3A4 inhibitor, or the statin can be held for the duration of a clarithromycin course.